<u>Thm</u> The only connected subsets of $\mathbb{R}$ are the intervals.

<u>Pf</u>: Suppose that $\emptyset \neq A \subseteq \mathbb{R}$ is connected. Want: A is an interval. Set

$$\alpha = \begin{cases} \inf A & \text{if } A \text{ is bounded below} \\ -\infty & \text{otherwise} \end{cases}$$

$$\beta = \begin{cases} \sup A & \text{if } A \text{ is bounded above} \\ +\infty & \text{otherwise} \end{cases}$$

Claim: $(\alpha, \beta) \subseteq A$.

- It then follows that $A = (\alpha, \beta)$, $(\alpha, \beta]$, $[\alpha, \beta)$ or $[\alpha, \beta]$ by our choice of α, β, and so A is an interval.

Suppose not: $\exists\, c \in (\alpha, \beta) \setminus A$. Set

$$B = (-\infty, c) \cap A, \qquad C = (c, \infty) \cap A.$$

Then:

① $B \cup C = A \setminus \{c\} = A$, since $c \notin A$.

② $B \neq \emptyset$: since $c > \alpha = \inf A$

$C \neq \emptyset$: since $c < \beta = \sup A$

③ B and C are separated:

$$B \subseteq (-\infty, c) \implies \overline{B} \subseteq (-\infty, c]$$

$$\implies \overline{B} \cap C \subseteq (-\infty, c] \cap (c, \infty) = \emptyset$$

$$C \subseteq (c, \infty) \implies \overline{C} \subseteq [c, \infty)$$

$$\Rightarrow \quad B \cap \bar{C} \subseteq (-\infty, c) \cap [c, \infty) = \emptyset$$

So A is disconnected — a contradiction. $\square$

- In summary:

<u>Cor</u> Let A be a subset of a metric space (X, d). The following are equivalent:

① A is connected.

② The only sets that are both open and closed in A (with the induced metric) are $\emptyset$ and A.

If $X = \mathbb{R}$: ③ A is an interval.

<u>Ex</u> $\mathbb{R}$ is connected, since $\mathbb{R}$ is an interval. So the only subsets of $\mathbb{R}$ that are open and closed are $\emptyset$ and $\mathbb{R}$.

<u>Ex</u> $\mathbb{Q}$ is not an interval, so it must be disconnected. So ① and ② above must be false. (See HW11.)

Continuity (Copson Ch. 7, Rudin Ch. 4)

<u>Def</u> Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$, and $f: A \to Y$. We say:

① the <u>limit</u> of f as x approaches $x_0 \in \bar{A}$ is $y \in Y$ if: $\forall \varepsilon > 0, \ \exists \delta > 0$ s.t.

$$x \in A, \quad 0 < d_X(x, x_0) < \delta \quad \Rightarrow \quad d_Y(f(x), y) < \varepsilon.$$

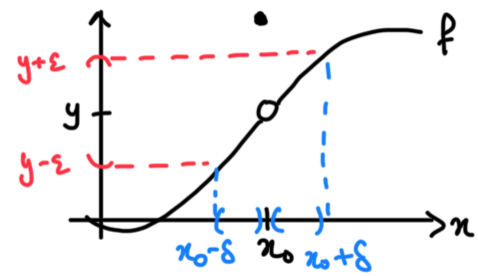
When this is true, we write "$\lim_{x \to x_0} f(x) = y$" or "$f(x) \to y$ as $x \to x_0$ in A".

② f is <u>continuous at x_0</u> if $x_0 \in A$ and $\lim_{x \to x_0} f(x) = f(x_0)$.

<u>Ex</u> ① For $f: \mathbb{R} \to \mathbb{R}$, this says:

$\forall \varepsilon > 0 \ \exists \delta > 0$ s.t.

$0 < |x - x_0| < \delta \Rightarrow |f(x) - y| < \varepsilon$.



- This is the usual definition (e.g. from Math 421)

Here, $\lim_{x \to x_0} f(x) = y \neq f(x_0)$.

② For $f: (a,b) \to \mathbb{R}$ and $x_0 = a$, this says:

$\forall \varepsilon > 0$, $\exists \delta > 0$ s.t.

$$x \in (a,b), \quad 0 < x - a < \delta \Rightarrow |f(x) - y| < \varepsilon.$$

This is the same as "$\lim_{x \to a^+} f(x) = y$".

<u>Prop</u> Let X, Y be metric spaces, $f: X \to Y$, and $x_0 \in X$. Then: f is continuous at x_0 $\Leftrightarrow$ $\forall$ sequence $\{x_n\}_{n \geq 1}$ in X that converges to x_0, the sequence $\{f(x_n)\}_{n \geq 1}$ in Y converges to $f(x_0)$.

<u>Pf</u>: $\boxed{\Rightarrow}$ Suppose $\lim_{x \to x_0} f(x) = f(x_0)$. Fix a sequence $\{x_n\}$ in X s.t. $x_n \to x$ as $n \to \infty$. Want: $f(x_n) \to f(x_0)$ in Y as $n \to \infty$. Fix $\varepsilon > 0$. As $\lim_{n \to \infty} f(x) = f(x_0)$, $\exists \delta > 0$ s.t.

$$x \in X, \quad \underbrace{\cancel{0 <}}_{} d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

Don't need: $d_X(x, x_0) = 0 \Rightarrow x = x_0$

$\Rightarrow d_Y(f(x_0), f(x_0)) = 0 < \varepsilon$

As $x_n \to x_0$ in X, $\exists N$ s.t.

$$d_X(x_n, x_0) < \delta \quad \forall n \geq N.$$

Together, for $n \geq N$ we have

$$d_X(x_n, x_0) < \delta \quad \Rightarrow \quad d_Y(f(x_n), f(x_0)) < \varepsilon.$$

So $f(x_n) \to f(x_0)$ in Y as $n \to \infty$.

$\boxed{\Leftarrow}$: Suppose that $\forall$ sequence $\{x_n\}$ in X that converges to x_0, $\{f(x_n)\}$ converges to $f(x_0)$ in Y. Suppose towards a contradiction that f is not continuous at x_0. Then $\exists \varepsilon > 0$ s.t. $\forall \delta > 0$,

$$0 < d_X(x, x_0) < \delta \not\Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

Taking $\delta = \frac{1}{n}$ for $n \in \mathbb{N}$, we get: $\exists x_n \in X$ s.t.

$$0 < d_X(x_n, x_0) < \frac{1}{n} \quad \text{and} \quad d_Y(f(x_n), f(x_0)) \geq \varepsilon.$$

Then $\{x_n\}$ is a sequence in X that converges to x_0, since $d_X(x_n, x_0) < \frac{1}{n} \to 0$ as $n \to \infty$. But $\{f(x_n)\}$ does not converge to $f(x_0)$ — a contradiction.

$\square$